

Problem

The imaginary part of impedance is called:

- (a) resistance
- (b) admittance
- (c) susceptance
- (d) conductance
- (e) reactance

Step-by-step solution

Step 1 of 3

The *impedance* $\mathbf{Z}$ of a circuit is the ratio of the phasor voltage, $\mathbf{V}$ to the phasor current, $\mathbf{I}$ measured in ohms (Ω).

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}}$$

The impedance is a complex quantity which is expressed in rectangular form as,

$$\mathbf{Z} = R + jX$$

Here, R is the real part of $\mathbf{Z}$ and it is known as *resistance*.

As the imaginary part of the impedance is not equal to resistance, option (a) is wrong.

X is the imaginary part of $\mathbf{Z}$ and it is known as *reactance*.

Step 2 of 3

The *admittance* $\mathbf{Y}$ of a circuit is the ratio of the phasor current to the phasor voltage of the circuit.

The *admittance* $\mathbf{Y}$ is the reciprocal of impedance, measured in Siemens ($\mathbf{S}$).

$$\begin{aligned}\mathbf{Y} &= \frac{1}{\mathbf{Z}} \\ &= \frac{\mathbf{I}}{\mathbf{V}}\end{aligned}$$

As the imaginary part of the impedance is not equal to admittance, option (b) is wrong.

Step 3 of 3

The *admittance* is a complex quantity which is expressed in rectangular form as,

$$\mathbf{Y} = G + jB$$

Here, G is the real part of $\mathbf{Y}$ and is known as *conductance*.

As the imaginary part of the impedance is not equal to conductance, option (d) is wrong.

B is the imaginary part of $\mathbf{Y}$ and it is known as *susceptance*.

As the imaginary part of the impedance is not equal to *susceptance*, option (c) is wrong.

Therefore, the imaginary part of $\mathbf{Z}$ is known as reactance.

Hence, the correct option is e.